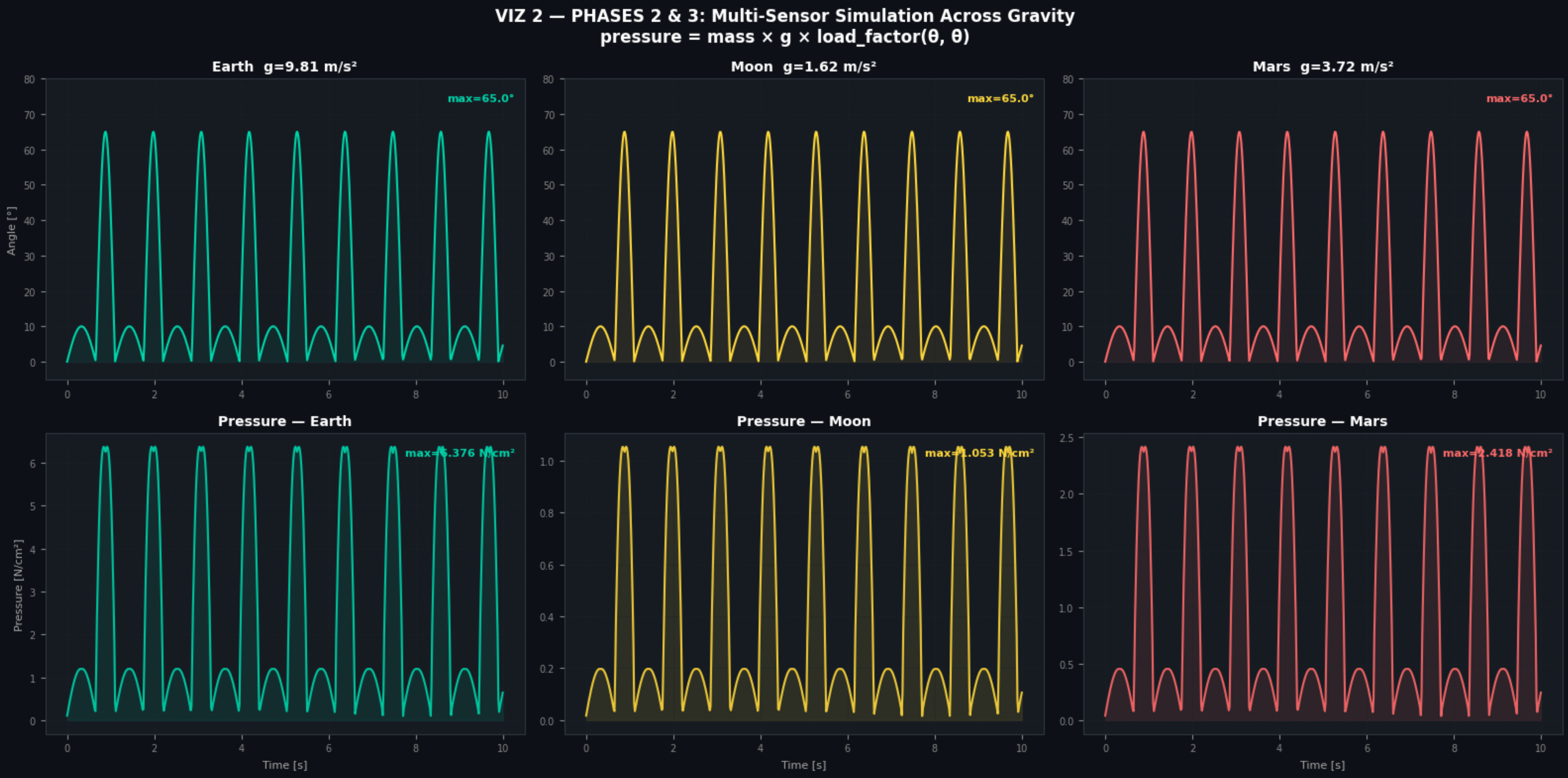
KNEE DIGITAL TWIN — V1

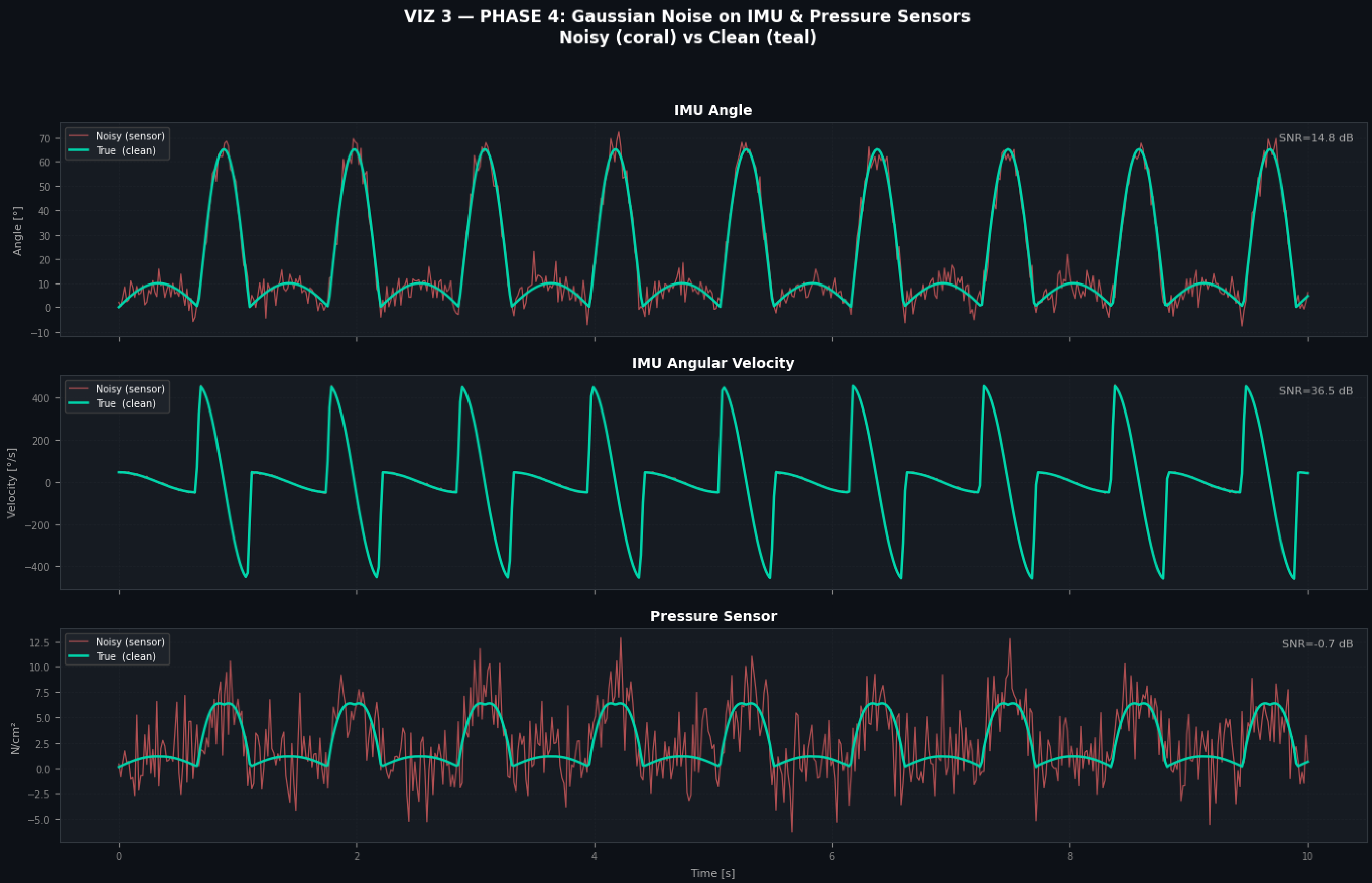
Adaptive Biomechanical Monitoring System

- Phase 1: Gait-Based Motion Modeling (piecewise, stance + swing)
- Phase 2-3: Multi-Sensor Simulation + Gravity Adaptation (Earth/Moon/Mars)
- Phase 4: Gaussian Noise Injection | Phase 5: Kalman Filter Estimation
- Phase 6: Biomechanical Feature Extraction | Phase 7: Risk Detection
- Phase 8: Visualization Dashboard + Animated GIF
- Total outputs: 9 Visualizations + 1 Animated GIF (key frames)

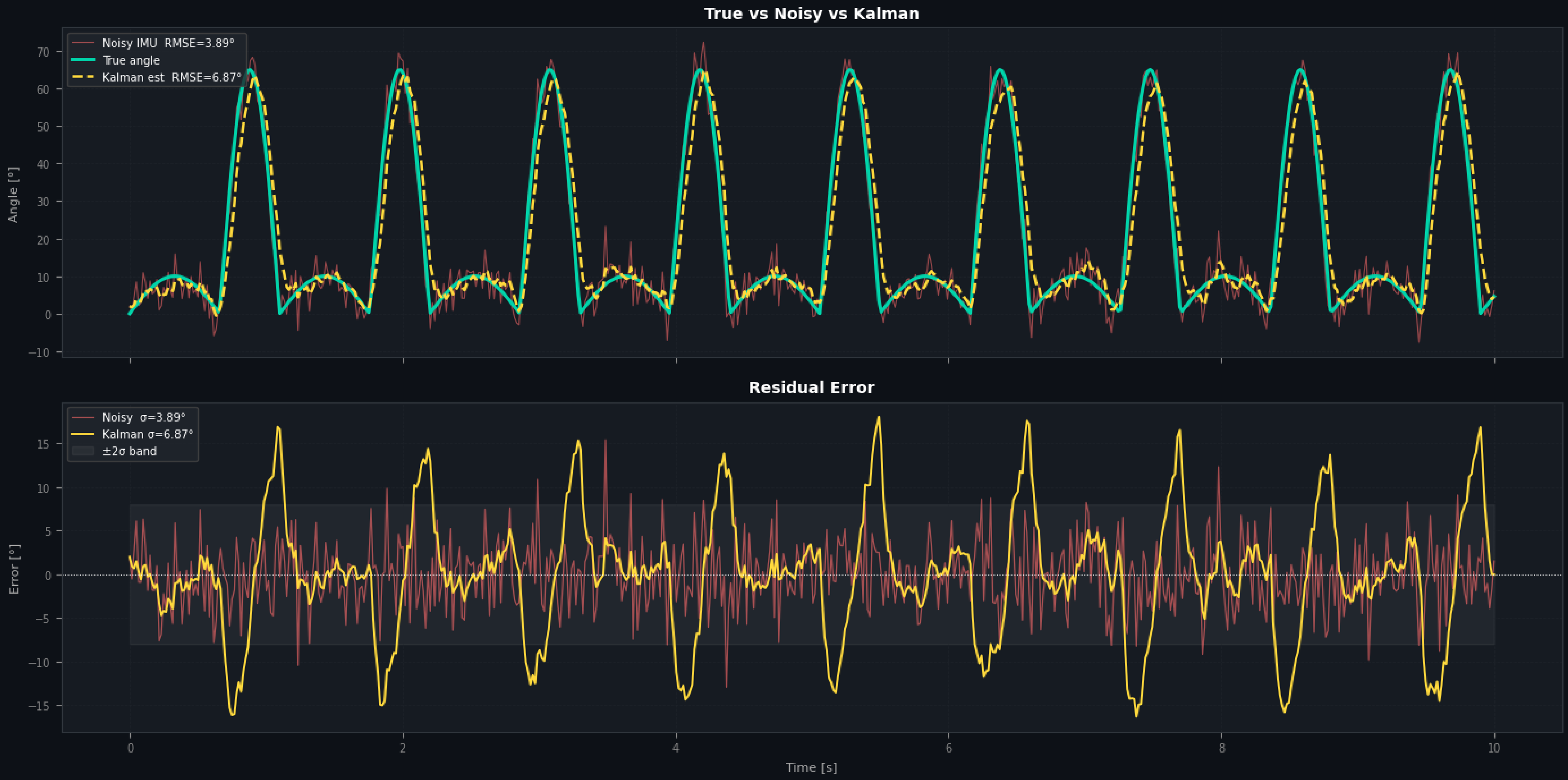
V1 OUTPUTS — TABLE OF CONTENTS

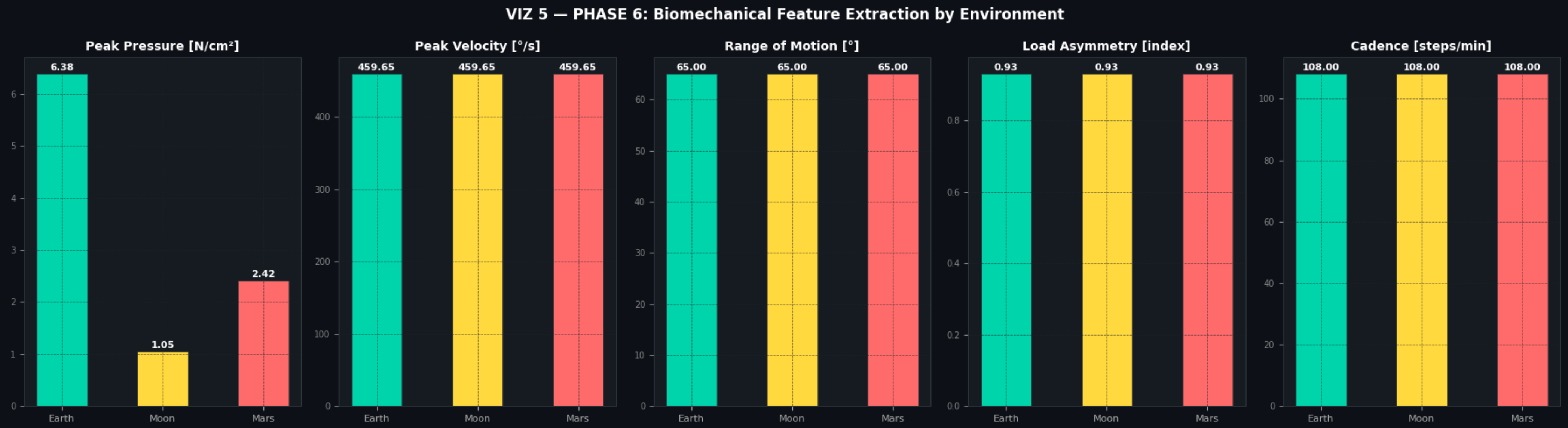

VIZ 1 — PHASE 1: Gait-Based Knee Motion Model
Blue band = Stance (60%) | Teal band = Swing (40%)
Flexion Angle
Stance 60%
Swing 40%
Angle [°]
Angular Velocity
Velocity [°/s]
Angular Acceleration
Accel [°/s²]
Time [s]


VIZ 2 — PHASES 2 & 3: Multi-Sensor Simulation Across Gravity
pressure = mass × g × load_factor(θ, θ)
Earth g=9.81 m/s²
Moon g=1.62 m/s²
Mars g=3.72 m/s²
max=65.0°
Angle [°]
Pressure — Earth
Pressure — Moon
Pressure — Mars
max=6.376 N/cm²
max=1.053 N/cm²
max=2.418 N/cm²
Pressure [N/cm²]
Time [s]


VIZ 3 — PHASE 4: Gaussian Noise on IMU & Pressure Sensors
Noisy (coral) vs Clean (teal)
IMU Angle
Noisy (sensor)
True (clean)
SNR=14.8 dB
Angle [°]
IMU Angular Velocity
SNR=36.5 dB
Velocity [°/s]
Pressure Sensor
SNR=-0.7 dB
N/cm²
Time [s]

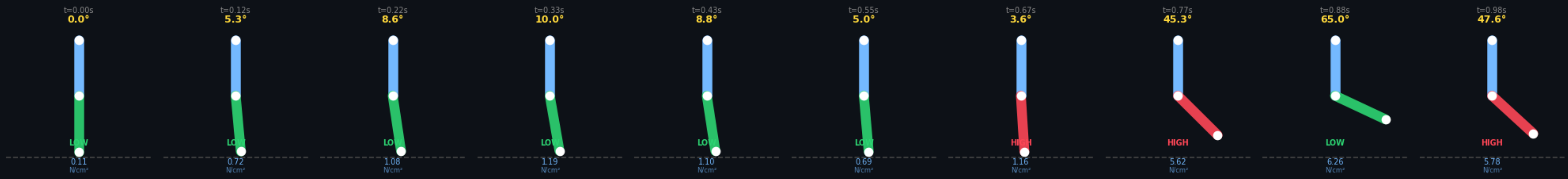
VIZ 4 — PHASE 5: Kalman Filter State Estimation
RMSE Noisy=3.89° → Kalman=6.87° (-76.9% improvement)

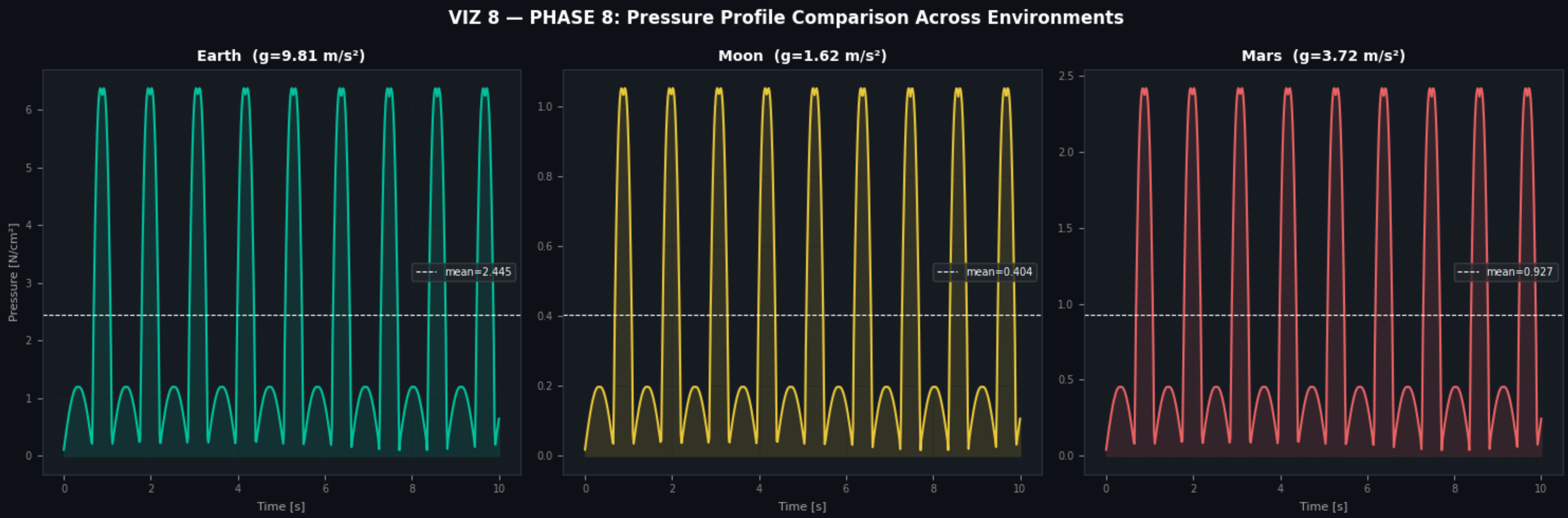



VIZ 5 — PHASE 6: Biomechanical Feature Extraction by Environment
Peak Pressure [N/cm²]
6.38
1.05
2.42
Peak Velocity [°/s]
459.65
459.65
459.65
Range of Motion [°]
65.00
65.00
65.00
Load Asymmetry [index]
0.93
0.93
0.93
Cadence [steps/min]
108.00
108.00
108.00
Earth
Moon
Mars

VIZ 6 — PHASE 7: Risk & Anomaly Detection Dashboard
RED = High | ORANGE = Medium | GREEN = Low

VIZ 7 — PHASE 8: Knee Anatomy Snapshots Across One Gait Cycle
Calf colour = risk | angle & pressure shown per frame

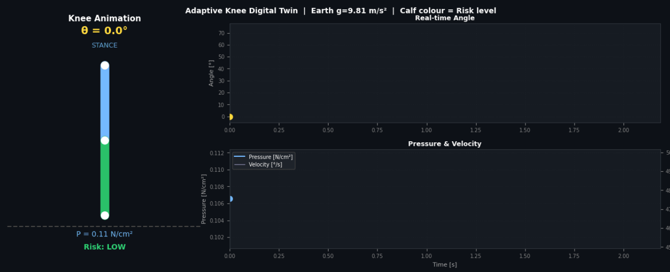



VIZ 8 — PHASE 8: Pressure Profile Comparison Across Environments
Earth (g=9.81 m/s²)
Moon (g=1.62 m/s²)
Mars (g=3.72 m/s²)
Pressure [N/cm²]
Time [s]
mean=2.445
mean=0.404
mean=0.927
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0
0.5
1.5
2.0
2.5
8
10

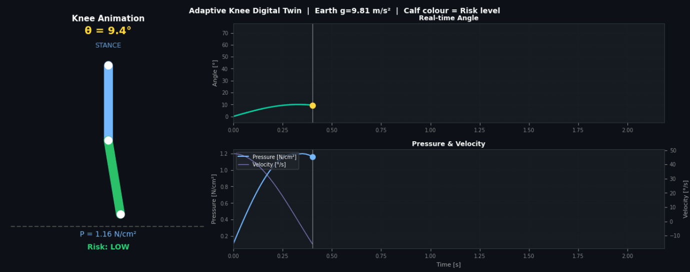

VIZ 9 — MASTER DASHBOARD: Adaptive Knee Digital Twin
Space Environments | All V1 Phases Integrated
Angle & Velocity
Angle [°]
Vel [°/s]
Kalman Filter
Noisy
True
Kalman
Angle [°]
Joint Pressure — All Environments
Earth g=9.81
Moon g=1.62
Mars g=3.72
N/cm²
Risk Classification
Limit 85.0°
HIGH
MEDIUM
LOW
|Angle| [°]
Velocity + Anomaly Zones
HIGH ±72.0
MED ±50.0
Vel [°/s]
Residual Error Analysis
Noisy σ=3.89°
Kalman σ=6.87°
Error [°]
Peak Pressure by Environment
6.376
1.053
2.418
N/cm²
Earth
Moon
Mars
Risk Distribution
High 37.0%
Medium 1.7%
Low 61.3%

GIF 1 — Live Knee Motion Animation (key frames)

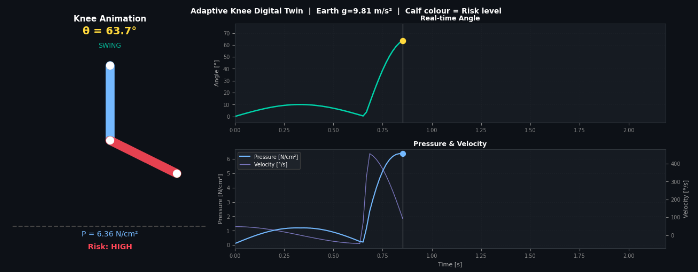
Animation Key Frames



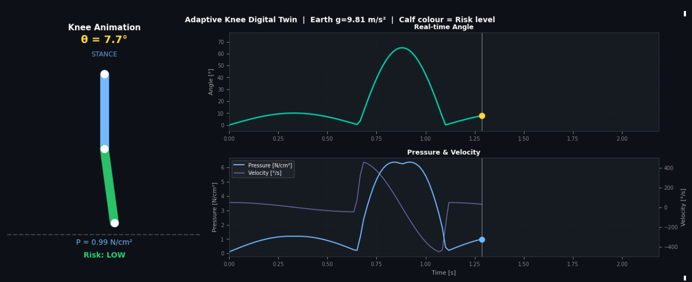
Frame 1



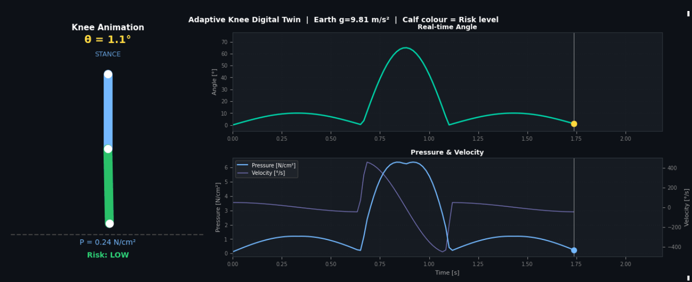
Frame 2



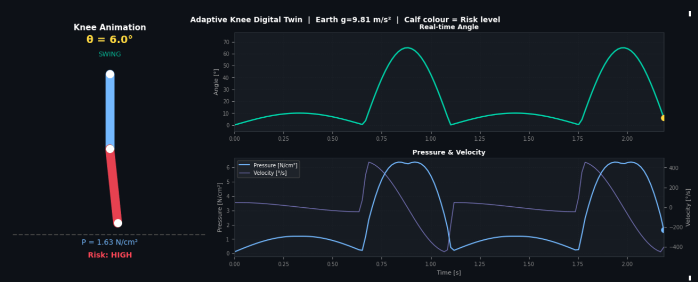
Frame 3



Frame 4



Frame 5



Frame 6